

THE EVOLUTION OF LLMs

From base model to agentic system

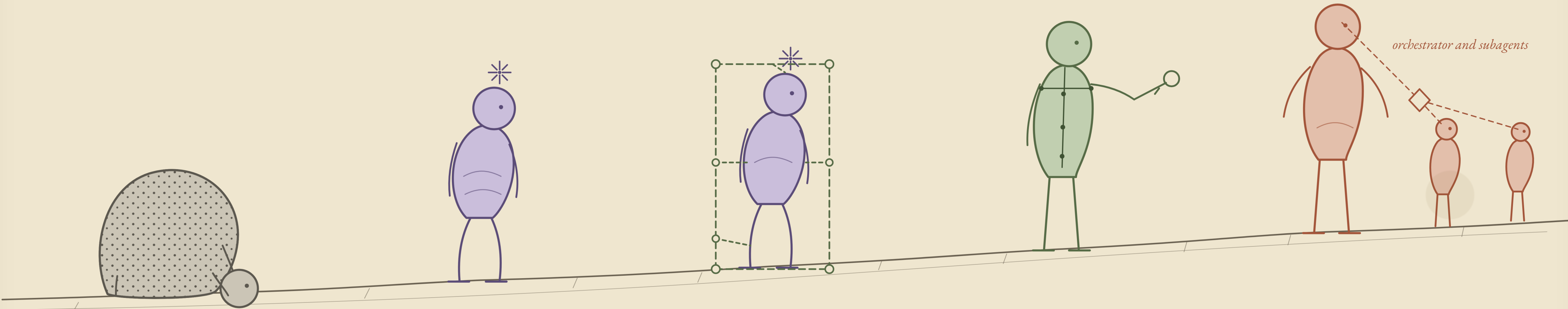
REINFORCEMENT LEARNING — THE SELECTIVE PRESSURE

c. 2022 · RLHF / DPO — rewards preferred answers (character)

c. 2024 · RLVR — verifiable rewards: reasoning emerges

c. 2025 · Agentic RL — trains the whole tool-action loop, not the single answer

Multi-agent RL (frontier)



I · PRE-TRAINING

Predicts the next token: the base model is born

GAP Knowledge frozen at the cutoff
REMEDY → RAG



II · POST-TRAINING

Alignment and character: from base to assistant (SFT, RLHF)

GAP Hallucinations, prompt dependence
REMEDY → Prompt engineering, self-reflection



III · HARNESS

Context, tools, execution loop: the scaffold around the model

GAP Limited context, token cost
REMEDY → Context management, telemetry and observability



IV · AI AGENT

Perceives, reasons, acts: ReAct loop and tool use

GAP Hallucinations extend to actions (wrong tool, invented parameters)
REMEDY → schemas, validators, permissions, human-in-the-loop



V · AGENTIC AI

Orchestrated specialized agents, shared memory

GAP Error cascades across agents, coordination
REMEDY → Role-based orchestration, monitoring/audit, governance



LEGEND

grey = raw knowledge purple = aligned model green = operational capabilities coral = multi-agent coordination

The notches count cumulative remedies: each stage inherits the fixes of the previous ones.